

WASP-2b

R-1116

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-2_141005 · created 2026-07-30T21:17:47+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456935.7513 +/- 0.0034 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.108 +/- 0.02
Transit depth (Rp/Rs)^2	1.16 +/- 0.44 [%]
Orbital Inclination (inc)	85.57 +/- 1.65
Airmass coefficient 1 (a1)	1.067 +/- 0.015
Airmass coefficient 2 (a2)	-0.0298 +/- 0.0025
Scatter in the residuals of the lightcurve fit is	1.42 %
Best Comparison Star	#1 - [67, 349]
Optimal Aperture	2.75
Optimal Annulus	4.02
Transit Duration (day)	0.074 +/- 0.019

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.108 ± 0.02 vs 0.1326 (deviation 1.15σ)
Duration fit vs published	0.074 d vs 0.0737 d (deviation 0.01σ)
Mid-time O–C	-5.4 min
Depth SNR	5.0
Residual scatter	1.42 %

Light curve

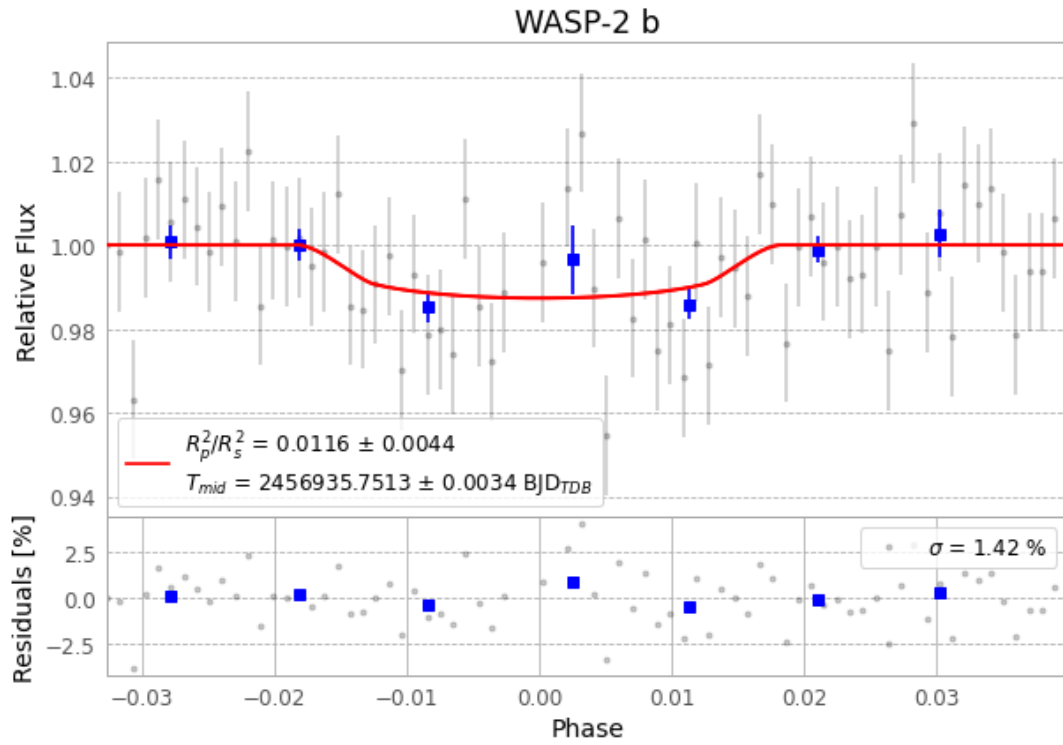


Figure 1 — FinalLightCurve_WASP-2 b_2014-10-04.png (SHA-256 f814f1122c3b9211...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [265, 291], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 907a3f3049a2c84c...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

76 FITS frames (50 MB), WASP-2141005041512.FITS → WASP-2141005080012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-2-141005.log (56672c34910f...), FinalLightCurve_WASP-2 b_2014-10-04.png (f814f1122c3b...), FinalLightCurve_WASP-2 b_2014-10-04.csv (1a89105c27f3...)

Compute resources

Wall time 120.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 21:17Z.